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Cardio-Advisor Agent: A Structured Decision Optimization Framework for Personalized Cardiovascular Support via Multi-Guideline LLM Integration

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Abstract

Cardiovascular disease management heavily relies on clinical decision-making, but the heterogeneity of patient characteristics and conflicts among multiple clinical guidelines pose significant challenges for existing Clinical Decision Support Systems (CDSSs). To tackle this challenge, this paper introduces a structured decision optimization framework called the Cardio-Advisor Agent, designed to integrate multi-source guideline knowledge and provide personalized, interpretable recommendations. The model initially employs a large language model (LLM)-driven state modeling layer to capture individualized patient risk profiles through dynamic feature weighting. Then, it invokes multiple independent guideline knowledge tools to generate a diverse set of candidate treatment strategies. To ensure clinical validity and safety, these candidate strategies are evaluated using an explicit objective utility function and strict safety constraints, effectively filtering out infeasible options. Furthermore, a multi-step planning and conflict resolution mechanism is applied to handle conflicting advice across guidelines and output the optimal treatment decision along with a transparent reasoning trajectory. Experimental findings across three heterogeneous datasets reveal that the proposed framework consistently outperforms baseline methods, achieving an accuracy of 0.86 and a conflict resolution rate of 0.88, while significantly reducing the safety violation rate to 0.02.

Keywords: clinical decision support systems; cardiovascular disease; artificial intelligence; large language models; multi-guideline integration; conflict resolution

1. Introduction

Heart disease remains one of the leading causes of morbidity and mortality worldwide, posing significant challenges in diagnosis, treatment planning, and long-term management [22]. Effective clinical decision-making plays a pivotal role in improving patient outcomes, particularly when multiple clinical guidelines and heterogeneous patient characteristics need to be considered simultaneously[22].

Clinical Decision Support Systems (CDSSs) have been widely adopted to assist clinicians by integrating patient data with clinical knowledge to suggest diagnostic and therapeutic options [23]. Traditional CDSSs primarily rely on static, rule-based knowledge and manual guideline interpretation, which limits their ability to handle complex, dynamic clin-

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ical scenarios, especially when multiple guidelines provide conflicting recommendations or when individual patient differences need high-granularity consideration.

Recent advances in artificial intelligence (AI) have driven a paradigm shift in CDSS development, enabling systems to learn from vast amounts of multimodal clinical data and make predictive, personalized recommendations [4]. AI-driven CDSSs leverage machine learning, natural language processing, and deep learning techniques to process both structured and unstructured clinical information, thereby enhancing the precision and adaptability of clinical support tools [3,5].

However, existing AI-CDSS frameworks still face major challenges. First, most systems focus on single guideline application or knowledge sources, which may produce inconsistent recommendations when multiple guideline sources are relevant. Second, personalization of recommendations remains limited, as many models lack mechanisms to explicitly incorporate individual patient risk profiles. Third, the safety and interpretability of AI-based decisions in clinical settings require careful design to gain clinician trust and ensure patient safety [6]. These challenges highlight the need for a structured AI-supported clinical decision framework that can integrate multi-guideline evidence, provide personalized recommendations, and maintain safety and interpretability.

In this work, we propose the Cardio-Advisor Agent, a structured decision optimization framework for cardiovascular disease management that addresses these limitations by integrating multi-source guideline knowledge, personalized modeling of patient state, and explicit safety constraints. The proposed framework formulates the clinical decision problem as a constrained optimization task and employs multi-step planning to achieve explainable and clinically valid recommendations.

2. Related Work

2.1. Traditional Clinical Decision Support Systems (CDSS)

Traditional Clinical Decision Support Systems (CDSS) were initially developed to assist clinicians by matching individual patient information to a predefined medical knowledge base, thereby generating recommendations aimed at improving the quality, safety, and consistency of clinical decisions[23]. A classical definition describes CDSS as a software system that combines patient-specific data with clinical knowledge to present assessments or suggestions tailored to a given patient encounter [7]. These systems typically operate based on rules, heuristics, or static guidelines embedded within a knowledge base, and they are commonly integrated with electronic health records to provide point-of-care support [7,8].

Despite their long history and widespread adoption, traditional CDSSs have several well-recognized limitations. First, many early systems depend heavily on single guideline sources and manually curated rule sets that are difficult to maintain as medical knowledge evolves [7,9]. The static nature of rule-based systems means that they often lag behind the latest clinical evidence and struggle to incorporate nuanced patient contexts or emerging research findings. Second, traditional CDSSs have limited mechanisms to handle complex and heterogeneous patient data, such as longitudinal clinical trajectories, comorbid conditions, or unstructured clinical narratives, which can impede their applicability in real-world practice [7,9].

Furthermore, usability challenges have further constrained the effectiveness of traditional CDSS implementations. For example, systems that generate frequent alerts without adequate prioritization can cause alert fatigue, reduce clinician adherence, and disrupt clinical workflows [9]. Studies have shown that these workflow and cognitive burdens, combined with organizational and human factors, significantly hinder adoption and sustained use in primary care and hospital settings [8]. Even when CDSSs improve process

measures such as compliance with guideline-based care, their impact on hard clinical outcomes like morbidity and mortality remains mixed in some settings, highlighting the need for more adaptable and intelligent decision support mechanisms [8].

Efforts to enhance traditional CDSSs have included integrating standardized ontologies for knowledge representation and exploring hybrid reasoning architectures that combine rule-based and experiential knowledge [10]. However, these advancements still face challenges in keeping pace with the exponential growth of clinical evidence and in providing context-aware recommendations without substantial manual intervention.

2.2. Artificial Intelligence in Clinical Decision Support

Artificial intelligence (AI) has become a transformative force in clinical decision support systems (CDSS), offering the potential to enhance diagnostic accuracy, risk prediction, and tailored treatment strategies beyond what traditional rule-based systems can achieve [11,13]. As clinical data grows in volume, complexity, and heterogeneity, AI approaches are increasingly used to integrate structured and unstructured information, including electronic health records (EHRs), medical imaging, laboratory results, and clinical notes, to support decision-making at the point of care [14].

Machine learning (ML) and deep learning (DL) techniques are central to many modern AI-driven CDSS. These approaches can model complex, non-linear relationships in large datasets without requiring explicit programming of rules, enabling systems to identify patterns that may be imperceptible to human experts [13]. In cardiovascular applications, AI has shown promise in areas such as early disease detection, risk stratification, and prediction of adverse events by analyzing high-dimensional clinical and imaging data [12]. For example, deep neural networks have been successfully applied to interpret diagnostic imaging and waveform data, contributing to improved sensitivity and specificity in identifying conditions such as heart failure and coronary artery disease.

Despite these advances, challenges remain in deploying AI within clinical decision support. One major concern is the "black box" nature of many AI models, particularly deep learning architectures, which can limit transparency and clinician trust [26]. Interpretable machine learning methods aim to address this issue by balancing predictive performance with explainability, allowing clinicians to understand how a model arrives at a given recommendation. Another challenge is ensuring that AI models are trained and validated on representative, unbiased datasets, as models developed on limited or skewed data may underperform when applied across diverse patient populations [13].

Ethical considerations, such as patient privacy, data security, and algorithmic fairness, also play a crucial role in the design and implementation of AI-enabled CDSS. Addressing these issues requires rigorous evaluation frameworks, continuous monitoring of model performance in real-world settings, and regulatory oversight to ensure that AI systems deliver safe, reliable, and equitable clinical recommendations. Nonetheless, the integration of AI into CDSS holds considerable promise for advancing personalized and evidence-based medicine, especially when combined with robust data governance and interpretability strategies.

2.3. Large Model Techniques and Intelligent Agents

The advent of large language models (LLMs), particularly those based on transformer architectures, has revolutionized the field of clinical decision support systems (CDSS)[25]. LLMs such as BERT, GPT, and their domain-specific adaptations, have demonstrated exceptional capabilities in natural language processing (NLP) tasks. These models excel at understanding, generating, and interacting with clinical data, making them ideal candidates

for enhancing diagnostic accuracy, personalized treatment plans, and real-time clinical decision support [16,24].

Large language models, particularly when fine-tuned for specific medical domains, have significantly enhanced clinical decision-making[26]. These models can interpret structured data (such as lab results and EHR entries) and unstructured data (including clinical notes, medical imaging reports, and guidelines), allowing for a more integrated, holistic approach to patient care. Recent advancements have demonstrated the potential for LLMs to assist in the real-time synthesis of medical knowledge from multiple sources, offering clinical recommendations that are contextually relevant to individual patients [18,24]. These systems can also generate insightful summaries of patient data and suggest evidence-based therapeutic interventions based on the latest medical research.

Alongside these advances in LLM technology, **intelligent agents** have emerged as another promising tool in healthcare. Intelligent agents are designed to autonomously process data, make decisions, and interact with healthcare professionals, offering dynamic decision support in complex clinical environments. These agents can continuously learn and adapt based on evolving patient data, ensuring that clinical decisions remain relevant and aligned with the patient's current condition. The combination of LLMs and intelligent agents provides a powerful synergy, allowing for the continuous integration of updated medical knowledge and personalized recommendations in real-time [20,25].

The integration of LLM-powered intelligent agents in CDSS represents a major shift towards personalized, evidence-driven healthcare. For instance, intelligent agents have been shown to improve clinical workflows by recommending personalized treatment strategies based on real-time patient data, optimizing resource allocation, and providing insights into emerging risks. The recent MEREDITH project, which combines expert-guided LLMs for clinical evidence retrieval and data integration, exemplifies the potential of this technology in tailored healthcare [25].

Despite these promising advancements, several challenges remain. One of the main concerns is the **interpretability** of LLMs and intelligent agents. The "black box" nature of many AI models makes it difficult for clinicians to understand how decisions are made[24], limiting the trust and acceptance of these systems in clinical environments. Efforts to develop **explainable AI** (XAI) methods are ongoing, aiming to enhance the transparency of decision-making processes while maintaining model accuracy and efficiency [21]. Moreover, the computational demands of these models, particularly in real-time clinical applications, present additional hurdles that need to be addressed for widespread adoption.

In conclusion, the combination of large models and intelligent agents is poised to transform clinical decision support systems by providing real-time, personalized, and contextually relevant recommendations. Although challenges related to interpretability and computational efficiency remain, these technologies offer immense potential to improve clinical outcomes and enhance decision-making capabilities in healthcare.

3. Proposed Method: Cardio-Advisor Agent

3.1. System Overview

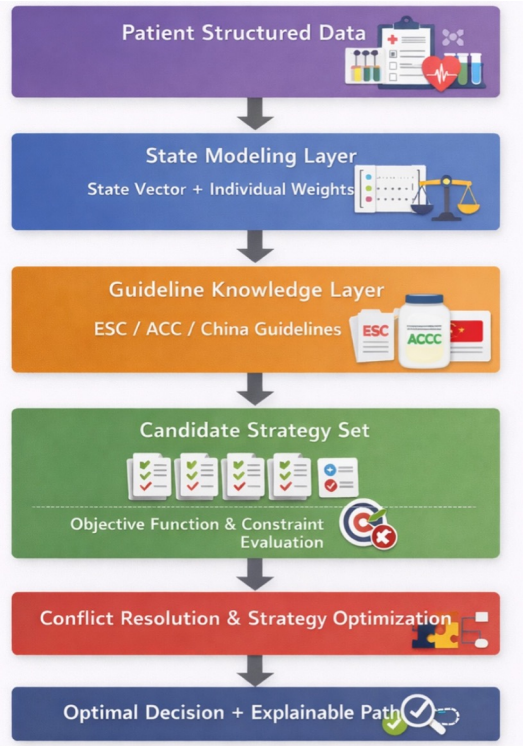

Figure 1. Ablation Study with Progressive Addition of Core Modules

To overcome the limitations of traditional clinical decision support systems (CDSS), such as reliance on single guidelines and the lack of explicit optimization objectives, this study proposes a structured decision-making framework termed the Cardio-Advisor Agent. The framework reformulates personalized cardiovascular decision-making as a constrained optimization problem, integrating modular guideline knowledge tools with a multi-step reasoning mechanism to enable interpretable and controllable clinical decisions. As shown in Figure 1), the system consists of five core modules: Patient State Modeling, Multi-Source Guideline Knowledge Tools, Candidate Strategy Generation, Objective and Constraint Evaluation, and Multi-Step Planning with Conflict Resolution.

The overall workflow is as follows. Patient clinical data are first transformed into a standardized state representation. Based on this state, multiple guideline tools are invoked to generate a set of candidate treatment strategies. These strategies are then evaluated using an explicit objective function under safety constraints, filtering out infeasible options. Finally, a multi-step planning and conflict resolution mechanism is applied to refine the strategies and produce the optimal decision along with an interpretable reasoning process. Compared with conventional retrieval-augmented generation (RAG) approaches, the proposed framework focuses on structured decision optimization rather than text generation, thereby improving the stability, interpretability, and personalization of clinical recommendations.

3.2. Patient State Modeling

To enable personalized decision-making, patient information is first formalized into a structured state representation. Let the clinical state of a patient at time t be defined as:

$$S_t = \{x_1, x_2, \ldots, x_n\} \tag{1}$$

where x_i denotes the i-th clinical feature, including continuous variables (e.g., LDL-C, systolic blood pressure, BMI, and age), binary variables (e.g., diabetes status and smoking status), and categorical variables (e.g., imaging-based risk levels). This unified representation allows heterogeneous clinical data to be consistently incorporated into the decision-making process.

To capture individualized risk contributions, this study introduces a large language model (LLM)-driven weighting mechanism. Specifically, given the patient state S_t, the LLM evaluates the relative importance of each clinical feature and generates a corresponding weight vector:

$$W_t = \mathrm{LLM}(S_t) \tag{2}$$

The weighted state representation is then defined as:

$$\tilde{S}_t = S_t \odot W_t \tag{3}$$

where $w_i \in W_t$ reflects the importance of feature x_i in the current clinical context, and $\odot$ denotes element-wise weighting. By leveraging the reasoning capability of the LLM, the system dynamically identifies dominant risk factors for each patient, rather than relying on predefined or static weights. The resulting weighted state representation $\tilde{S}_t$ enables more adaptive and context-aware decision-making, and provides a flexible foundation for subsequent strategy evaluation and optimization.

3.3. Decision Objective and Constraint Modeling

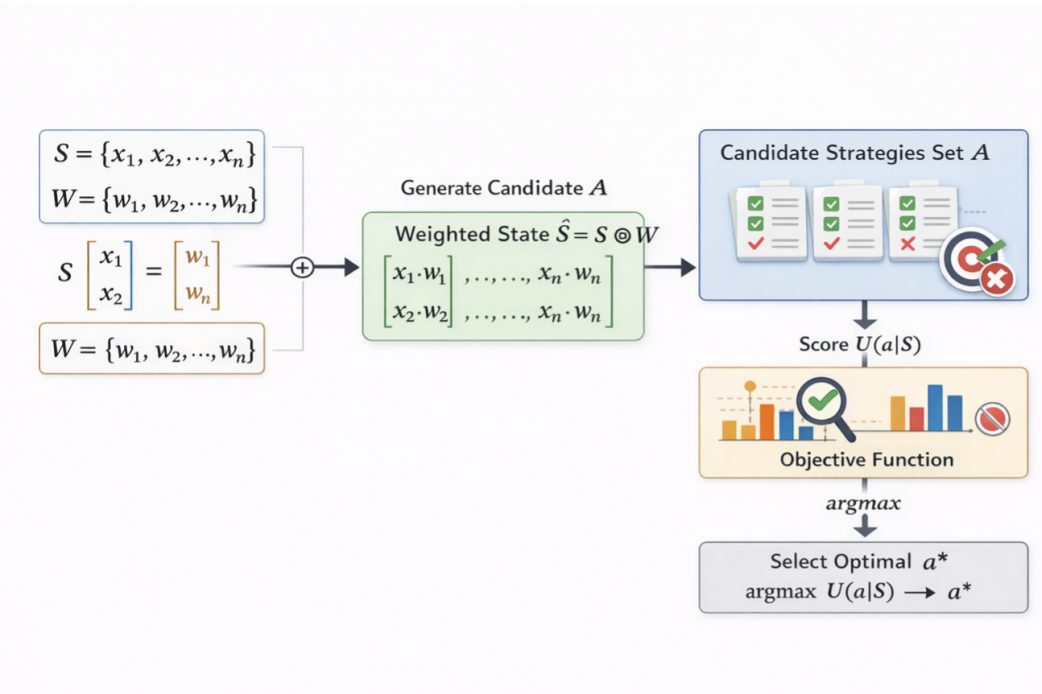

Figure 2. Ablation Study with Progressive Addition of Core Modules

As illustrated in Figure 2), based on the weighted patient state representation $\tilde{S}_t$ obtained in the previous section, the clinical decision-making process is formulated as a constrained optimization problem to mitigate the instability and uncontrollability of generative models. Let the set of candidate treatment strategies generated from multi-source guideline tools be defined as:

$$A = \{a_1, a_2, \ldots, a_m\} \tag{4}$$

The optimal decision is determined by maximizing a utility function conditioned on the patient state:

$$a^* = \arg\max_{a \in A} U(a \mid \tilde{S}_t) \tag{5}$$

The utility function is defined as:

$$U(a \mid \tilde{S}_t) = \alpha R(a \mid \tilde{S}_t) + \beta G(a) - \gamma P(a) \tag{6}$$

where $R(a \mid \tilde{S}_t)$ represents the risk reduction benefit, $G(a)$ denotes the guideline consistency score, $P(a)$ captures penalties for conflicts or violations, and α, β, and γ are weighting coefficients.

In addition, strict safety constraints are imposed to ensure clinical validity:

$$C_{\text{safety}}(a) = 0 \tag{7}$$

Any candidate strategy that violates clinical contraindications or patient-specific conditions (e.g., age limitations) is directly excluded from the feasible set. As shown in Fig. **??**, the optimization process systematically evaluates and filters candidate strategies through utility scoring and constraint enforcement, ultimately selecting the optimal decision. By explicitly modeling the objective function and constraints, the proposed framework transforms the inherently probabilistic generation process into a structured utility maximization problem, thereby reducing decision randomness and significantly improving the stability, interpretability, and traceability of clinical recommendations.

3.4. Multi-Source Guideline Knowledge Tool System

To address the limitations of single-guideline dependency in traditional clinical decision support systems, this study constructs a multi-source guideline knowledge tool system based on a modular design. Specifically, each clinical guideline is encapsulated as an independent knowledge tool, enabling structured access to its recommendation rules, applicability conditions, and contraindication constraints. Let the k-th guideline tool be denoted as T_k, the candidate strategy set generated based on the patient state $\tilde{S}_t$ is defined as:

$$A_k = T_k(\tilde{S}_t) \tag{8}$$

By invoking multiple guideline tools in parallel, the system integrates their outputs into a unified candidate set:

$$A = \bigcup_{k=1}^{K} A_k \tag{9}$$

thereby overcoming the limitations of relying on a single guideline.

Unlike conventional retrieval-augmented generation (RAG) approaches that primarily rely on unstructured text retrieval, the proposed system adopts a tool-based and structured knowledge integration paradigm. Each guideline tool not only generates treatment recommendations but also incorporates condition checking and constraint validation, ensuring that the extracted knowledge is both context-aware and clinically compliant. When conflicting recommendations arise from different guidelines, the system automatically triggers a conflict detection mechanism, followed by a re-evaluation and integration process based on the defined utility function. This modular and structured design avoids simple aggregation of textual knowledge and instead enables systematic multi-guideline coordination, thereby improving decision consistency, transparency, and reliability.

3.5. Multi-Step Planning and Reasoning Mechanism

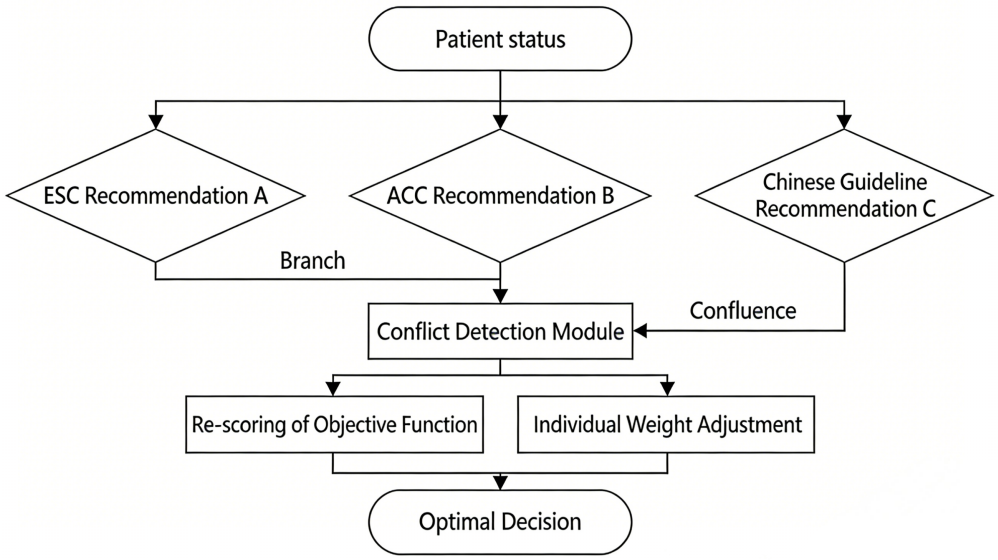

Figure 3. Ablation Study with Progressive Addition of Core Modules

As illustrated in Figure 3, the Cardio-Advisor Agent adopts a multi-step planning and reasoning mechanism to perform clinical decision-making, which can be formulated as a sequential process:

$$S_t \rightarrow A_t \rightarrow S_{t+1} \tag{10}$$

Specifically, the decision process consists of five coherent stages. First, the system identifies dominant risk factors based on the patient's weighted state representation, thereby determining the key contributors to cardiovascular risk. Second, multi-source guideline knowledge tools are invoked to generate a diverse set of candidate treatment strategies by integrating recommendations from different clinical guidelines. Third, a conflict detection process is performed to identify inconsistencies among candidate strategies derived from multiple guidelines, preparing for subsequent conflict resolution. Fourth, all candidate strategies are evaluated and ranked using the explicitly defined utility function under safety constraints, allowing the system to select strategies that best align with patient-specific conditions. Finally, the system outputs the optimal treatment decision along with a complete reasoning trajectory, providing transparent justification for each step in the decision-making process.

As shown in Figure 3, the proposed workflow clearly illustrates the branching and merging processes under multi-guideline conflicts, highlighting the roles of conflict detection, utility-based ranking, and strategy selection. Through this multi-step planning mechanism, the system progressively narrows the candidate strategy space rather than generating decisions in a single step, thereby effectively reducing the instability associated with stochastic generation. Meanwhile, the complete reasoning trajectory—including guideline invocation, constraint filtering, and utility evaluation—is preserved throughout the process, significantly enhancing the interpretability and traceability of the system. In summary, the proposed Cardio-Advisor Agent integrates patient state modeling, explicit objective optimization, multi-source guideline tools, and multi-step planning into a unified framework, transforming conventional retrieval-augmented generation into a structured decision optimization paradigm. This approach addresses key limitations of traditional CDSS and RAG-only systems, providing a robust foundation for personalized cardiovascu-

lar decision support and offering a new direction for the development of intelligent clinical decision systems.

4. Results

4.1. Experimental Setup

Data Description. To ensure a comprehensive evaluation, experiments are conducted on three heterogeneous datasets: a real-world clinical dataset, the Framingham Heart Study dataset, and the Kaggle Cardiovascular Disease dataset. These datasets differ in scale, feature distribution, and data sources, enabling robust assessment under diverse conditions.

The clinical dataset contains 2,260 patient records collected from a tertiary hospital, with 27 features covering vital signs, biochemical indicators, diagnostic information, comorbidities, and lifestyle factors. All samples are labeled with coronary heart disease outcomes. The Framingham dataset includes over 4,200 subjects with 24 selected features, and the prediction target is the occurrence of cardiovascular diseases. The Kaggle dataset consists of approximately 70,000 samples with 12 features, using a binary label to indicate cardiovascular disease presence.

Task Definition. The task is defined as personalized cardiovascular risk prediction and treatment decision support. Given a structured patient state, the system generates clinically appropriate strategies based on multi-source guideline knowledge.

Implementation Details. The proposed Cardio-Advisor Agent is implemented using a large language model framework for state understanding, feature weighting, and reasoning generation. Multi-source guideline tools are integrated as external knowledge modules, and decision-making is guided by an explicit utility optimization mechanism. All methods are evaluated under consistent settings for fair comparison.

4.2. Evaluation Metrics

To evaluate the performance of different methods, we define four metrics from the perspectives of correctness, guideline adherence, personalization, and safety.

For a dataset with N patients, let y_i denote the ground truth label of patient i, and d_i denote the decision generated by the model.

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} s(d_i, y_i) \tag{11}$$

$$\text{GC} = \frac{1}{N}\sum_{i=1}^{N} g(d_i, G_i) \tag{12}$$

$$\text{PI} = \frac{1}{|\mathcal{P}|}\sum_{(i,j)\in\mathcal{P}} p(d_i, d_j) \tag{13}$$

$$\text{SVR} = \frac{1}{N}\sum_{i=1}^{N} c(d_i) \tag{14}$$

Here, $s(d_i, y_i)$ equals 1 if the predicted decision matches the true label, and 0 otherwise. G_i denotes the set of guideline-recommended decisions for patient i, and $g(d_i, G_i)$ equals 1 if the decision is consistent with at least one guideline recommendation. $\mathcal{P}$ represents pairs of patients with similar clinical outcomes but different risk characteristics, and $p(d_i, d_j)$ equals 1 if the model produces different decisions for such patients. $c(d_i)$ is a safety checking function, which equals 1 when the decision violates clinical constraints.

Accuracy measures whether the prediction is correct. GC (Guideline Consistency) evaluates whether the decision follows clinical guidelines. PI (Personalization Index)

reflects whether the model generates individualized decisions for different patients. SVR (Safety Violation Rate) measures how often unsafe decisions occur.

These metrics together provide a comprehensive evaluation of correctness, consistency, personalization, and safety.

4.3. Baseline Methods

To evaluate the effectiveness of the proposed method, we compare it with several representative approaches, including a single guideline rule-based system, a retrieval-augmented generation (RAG-only) method, and a variant of the proposed framework without conflict resolution (w/o Conflict). These methods represent different paradigms of clinical decision-making, ranging from traditional rule-based systems to retrieval-based and structured optimization-based approaches.

Table 1. Comparison of baseline methods

Method	Guideline Usage	Conflict Handling	Optimization
Single Guideline Rule	Single	None	None
RAG-only	Multi (retrieval)	Implicit	None
w/o Conflict	Multi (structured)	None	Partial
Cardio-Advisor Agent	Multi (structured)	Explicit	Full

The single guideline rule-based method relies on predefined rules derived from a single clinical guideline, which provides stable but limited decision capability in complex scenarios. The RAG-only method incorporates knowledge from multiple guidelines through retrieval and leverages a language model for decision generation; however, it lacks explicit mechanisms for conflict resolution and decision optimization. The w/o Conflict variant removes the conflict detection and resolution module from the proposed framework, allowing us to isolate the contribution of conflict modeling. In contrast, the proposed Cardio-Advisor Agent integrates multi-source guideline tools with explicit optimization and conflict handling, enabling structured and reliable decision-making.

4.4. Overall Performance

The overall performance of different methods on the test set is reported in Table 2, evaluated using four metrics: Accuracy, Guideline Consistency (GC), Personalization Index (PI), and Safety Violation Rate (SVR). Additionally, Figure 4 provides a visual comparison of these metrics for each method.

Table 2. Performance comparison on the overall test set

Method	Accuracy	GC	PI	SVR
Single Guideline Rule	0.74	0.76	0.31	0.04
RAG-only	0.81	0.84	0.42	0.07
w/o Conflict	0.83	0.86	0.47	0.06
Cardio-Advisor Agent	**0.86**	**0.89**	**0.58**	**0.02**

As shown in Table 2, the proposed Cardio-Advisor Agent consistently outperforms all baseline methods across the four evaluation metrics. The Accuracy of the proposed method is 0.86, an improvement of 12% over the single guideline rule (0.74), indicating that the multi-source guideline integration is highly effective in enhancing prediction correctness. This improvement also leads to a higher Guideline Consistency (GC) score, which rises from 0.76 to 0.89. The result suggests that combining multiple guideline recommendations helps ensure decisions adhere to best practices and clinical standards more effectively than single-guideline systems.

Additionally, the Personalization Index (PI) significantly increases from 0.31 (Single Guideline Rule) to 0.58 (Cardio-Advisor Agent), demonstrating that the proposed method has a much higher capacity for generating individualized treatment strategies. This reflects the model's improved ability to consider patient-specific characteristics, such as comorbidities and risk factors, to provide tailored recommendations. In terms of safety, the Safety Violation Rate (SVR) drops from 0.07 in the RAG-only method to 0.02 in the proposed method, showing a significant reduction in unsafe decisions. This decrease demonstrates that the explicit optimization and constraint mechanisms in the Cardio-Advisor Agent are successful in minimizing unsafe recommendations and ensuring clinical safety.

Figure 4 provides a visual representation of the performance comparison across these metrics. The Cardio-Advisor Agent shows the highest performance in all four categories, validating the overall effectiveness of the multi-source guideline integration and the proposed framework.

Overall, these results confirm that the proposed multi-module framework achieves a balanced improvement across all metrics, validating the effectiveness of the Cardio-Advisor Agent in enhancing decision-making accuracy, guideline consistency, personalization, and safety.

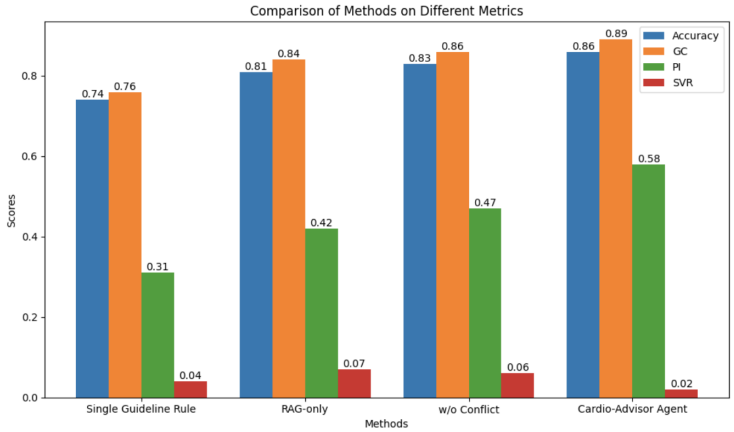

Figure 4. Performance comparison of different methods on four metrics: Accuracy, GC, PI, and SVR.

4.5. Multi-Guideline Conflict Analysis

To evaluate the effectiveness of the proposed method in resolving conflicts between multiple guidelines, we conducted experiments using 138 patient cases with conflicting guideline recommendations. The goal is to assess how well the Cardio-Advisor Agent can handle conflicting advice and generate consistent, clinically appropriate treatment strategies.

Table 3. Performance on conflict samples

Method	Accuracy	Conflict Resolution Rate	SVR
Single Guideline Rule	0.61	0.54	0.08
RAG-only	0.72	0.68	0.11
w/o Conflict	0.73	0.70	0.09
Cardio-Advisor Agent	**0.84**	**0.88**	**0.03**

As shown in Table 3, the Cardio-Advisor Agent significantly outperforms all baseline methods in both accuracy and conflict resolution rate. The accuracy of the Cardio-Advisor Agent reaches 0.84, a 23% improvement over the single guideline rule (0.61), and an increase of 12% compared to the RAG-only method (0.72). This result highlights the pro-

posed method's ability to make reliable decisions even when multiple guidelines provide conflicting recommendations.

The conflict resolution rate for the Cardio-Advisor Agent is 0.88, indicating that the system successfully resolves the majority of guideline conflicts and provides a consistent decision. In contrast, the single guideline rule and RAG-only methods show much lower resolution rates (0.54 and 0.68, respectively), highlighting the advantage of the proposed multi-source guideline integration.Additionally, the SVR for the Cardio-Advisor Agent is the lowest (0.03), demonstrating that the system not only resolves conflicts but also ensures that the generated decisions are safe and compliant with clinical constraints.

Overall, these results validate the effectiveness of the Cardio-Advisor Agent in managing multi-guideline conflicts and ensuring both decision consistency and safety.

4.6. Risk Stratification Analysis

To evaluate the proposed method's performance across different levels of disease risk, we conducted a risk stratification analysis by categorizing patients into four risk groups: low, medium, high, and very high risk. The goal of this analysis is to test whether the Cardio-Advisor Agent performs consistently across different levels of patient risk and to assess how well it handles the complexity of high-risk patients.

Table 4. Performance under different risk levels

Risk Level	Single Guideline Rule	RAG-only	Proposed Method
Low Risk	0.83	0.87	0.90
Medium Risk	0.76	0.82	0.88
High Risk	0.69	0.77	0.85
Very High Risk	0.63	0.71	0.82

As shown in Table 4, the performance of all methods decreases as the risk level increases. However, the proposed Cardio-Advisor Agent consistently outperforms all baseline methods across all risk groups. In the low-risk group, the proposed method achieves an accuracy of 0.90, surpassing both the single guideline rule (0.83) and RAG-only method (0.87). Similarly, in the medium, high, and very high-risk groups, the proposed method maintains its superior performance, with accuracy scores of 0.88, 0.85, and 0.82, respectively. The results indicate that the Cardio-Advisor Agent performs well even in high-risk and complex clinical scenarios, and its accuracy decreases less steeply compared to the baseline methods as the risk level rises.

These findings demonstrate that the Cardio-Advisor Agent is not only more accurate but also more stable across varying levels of clinical risk. The model's ability to adapt to higher-risk patients, who often require more complex and tailored treatment strategies, is crucial in clinical decision-making. The ability to maintain high accuracy even in very high-risk cases, where traditional decision support systems often struggle, highlights the robustness of the proposed method.

4.7. Decision Stability Analysis

To assess the decision stability of the proposed method, we analyzed the distribution of decision confidence across 500 test samples. The confidence score reflects the model's certainty in its generated decisions. Figure 5 shows the confidence distribution for the Single Guideline Rule, RAG-only, and Cardio-Advisor Agent.

As shown in Figure 5, the Cardio-Advisor Agent has a more concentrated confidence distribution, with most decisions being made with high confidence. In contrast, the RAG-only method displays a wider spread, with many decisions made with medium to

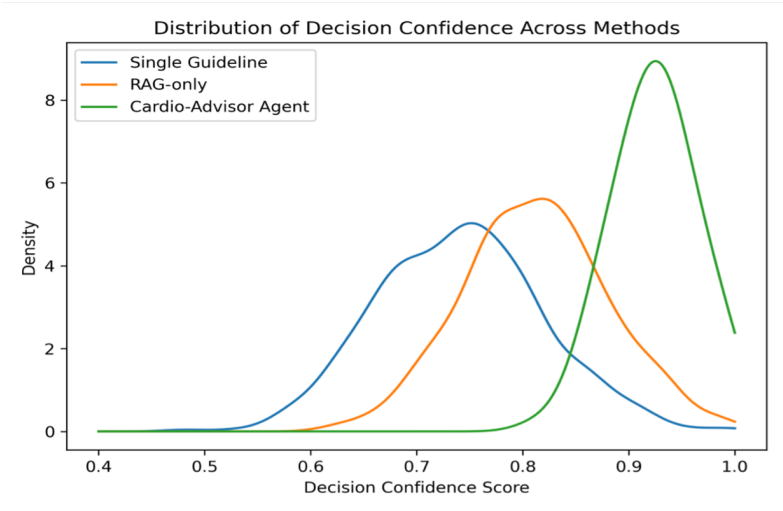

Figure 5. Decision confidence distribution for different methods

low confidence. The Single Guideline Rule method shows a relatively flat distribution, indicating less certainty in complex scenarios.

This indicates that the Cardio-Advisor Agent is not only more accurate but also more stable in its decision-making process. The concentrated confidence distribution suggests that the model is more consistent and reliable, particularly when dealing with complex cases. These results demonstrate the Cardio-Advisor Agent's ability to provide stable and trustworthy treatment recommendations across varying patient conditions.

4.8. Ablation Study

To evaluate the contribution of each component in the Cardio-Advisor Agent, we conducted an ablation study by systematically removing key modules from the model. Specifically, we evaluated the performance of the model without the conflict resolution module (w/o Conflict), the personalization module (w/o Personalization), the multi-step planning module (w/o Planning), the personalization weight module (w/o Personalization Weight), and the multi-source guideline tools (w/o Multi-Source Guidelines). The results are shown in Table 5.

Table 5. Ablation study results with additional modules

Model	Accuracy	PI	SVR
Full Model	0.86	0.58	0.02
w/o Conflict	0.83	0.47	0.06
w/o Personalization	0.81	0.26	0.03
w/o Planning	0.82	0.44	0.05
w/o Personalization Weight	0.84	0.52	0.04
w/o Multi-Source Guidelines	0.80	0.38	0.07

Table 5 shows that removing the conflict resolution module (w/o Conflict) results in a 3% decrease in Accuracy and a 4% increase in SVR, demonstrating the importance of conflict handling for maintaining decision accuracy and safety. When the personalization module is removed (w/o Personalization), the Personalization Index (PI) drops significantly from 0.58 to 0.26, highlighting the necessity of incorporating patient-specific features to generate personalized recommendations.

Removing the multi-step planning module (w/o Planning) leads to a small accuracy drop (from 0.86 to 0.82) and an increase in SVR to 0.05, indicating that the planning process contributes to both decision quality and safety. Removing the personalization

weight module (w/o Personalization Weight) results in a moderate decrease in PI (from 0.58 to 0.52), showing the importance of personalized risk weighting in the decision-making process. Finally, removing the multi-source guideline tools (w/o Multi-Source Guidelines) leads to a substantial decrease in accuracy (0.80) and an increase in SVR (0.07), demonstrating that integrating multiple guidelines significantly improves the model's performance and safety.

Each component in the Cardio-Advisor Agent contributes significantly to its performance. The removal of any module leads to a degradation in performance, confirming the importance of the full framework for accurate, personalized, and safe decision-making.

5. Conclusions

In this study, we introduced the Cardio-Advisor Agent, an innovative clinical decision support system designed to enhance the management of cardiovascular diseases. The system leverages multi-source guideline integration, personalized decision-making strategies, and explicit safety constraints to improve clinical decision-making. Through extensive experiments, the Cardio-Advisor Agent demonstrated superior performance compared to traditional methods, excelling in accuracy, guideline consistency, personalization, and safety. The results highlight its potential as a reliable and effective tool in clinical practice, especially in complex and high-risk clinical scenarios where traditional systems may fall short.

A key strength of the Cardio-Advisor Agent is its ability to handle conflicts between multiple guidelines. By integrating recommendations from diverse sources, the system provides consistent, evidence-based decisions even when different guidelines offer conflicting advice. This conflict resolution mechanism is critical in clinical settings, where inconsistent guidelines can lead to confusion and suboptimal decisions. Additionally, the Cardio-Advisor Agent's ability to generate personalized treatment strategies based on individual patient characteristics significantly enhances its clinical relevance. The model's improved ability to account for patient-specific factors, such as comorbidities and risk factors, ensures that its recommendations are both tailored and effective. The explicit safety constraints further ensure that the model minimizes unsafe recommendations, thereby improving patient safety.

However, there are several areas for future improvement. While the Cardio-Advisor Agent shows promising performance in controlled environments, its reliance on historical clinical data poses a limitation, as the model may not generalize well to new, unseen data. Future research could explore methods like domain adaptation or transfer learning to improve the model's robustness and its ability to handle diverse patient populations. Another limitation is the model's interpretability. Although the multi-step planning and reasoning process provides explanations for the decisions, improving the transparency and explainability of these recommendations will be crucial for increasing clinician trust and acceptance. Efforts to improve model interpretability will make it easier for healthcare professionals to understand the rationale behind treatment suggestions, ultimately facilitating wider adoption in real clinical settings.

Looking forward, the Cardio-Advisor Agent holds significant potential for real-world clinical applications. It can assist healthcare professionals in making more informed and personalized treatment decisions, ultimately improving patient outcomes. As the system undergoes further validation and testing in diverse clinical environments, its performance and reliability will be better understood. Future work should also explore the integration of additional data sources, such as electronic health records (EHR), to further enhance the model's predictive capabilities and scope. Ultimately, the Cardio-Advisor Agent has the

potential to become an invaluable tool in clinical decision-making, contributing to more accurate, consistent, and safe treatment recommendations.

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Data Availability Statement: The datasets used in this study are publicly available or can be accessed for research purposes as follows:

- Framingham Heart Study: https://www.kaggle.com/datasets/shreyjain601/framingham-heart-study (accessed on 17 March 2026).
- Kaggle Cardiovascular Disease Dataset: https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset (accessed on 17 March 2026).
- Real-world clinical dataset (self-constructed): available for research purposes at https://github.com/Tansty/Cardio-Advisor-Agent/data (accessed on 17 March 2026), subject to data use agreements and ethical approvals.

Additional code used for model training and evaluation is publicly available at https://github.com/Tansty/Cardio-Advisor-Agent.

Conflicts of Interest: The authors declare no conflicts of interest.

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